1- **CLASS TASKS SOLUTIONS**

--- TASK 1, 2, 3 & 4 SOLUTION KEY (PYTHON CODE) ---

```Python

import pandas as pd
import numpy as np

# Task 1: Setup & Initial Defect Audit
raw_returns_data = {
    "return_id": ["RET-301", "RET-302", "RET-303", "RET-303", "RET-304",
"RET-305", "RET-306", "RET-307"],
    "log_date": ["2026-10-01", "2026-10-01", "2026-10-02", "2026-10-02",
"2026-10-03", "2026-10-03", "2026-10-04", "2026-10-04"],
    "item_category": [" APPAREL ", "Electronics", " home ", " home ",
"Apparel", " ELECTRONICS ", "Apparel", "Home"],
    "refund_usd": [45.00, 299.00, np.nan, np.nan, 60.00, 150.00, np.nan,
85.00],
    "customer_rating": [1, 2, 1, 1, 3, 2, 1, 4]
}

df_returns = pd.DataFrame(raw_returns_data)

print("Initial Shape:", df_returns.shape) # (8, 5)
print("\nMissing Values:\n", df_returns.isnull().sum()) # refund_usd: 3
print("Duplicate Rows Count:", df_returns.duplicated().sum()) # 1

# Task 2: Deduplication & String Cleaning
df_returns = df_returns.drop_duplicates().reset_index(drop=True)
df_returns["item_category"] =
df_returns["item_category"].str.strip().str.title()

print("\nUnique Categories:", df_returns["item_category"].unique()) #
['Apparel', 'Electronics', 'Home']

# Task 3: Date Parsing & Imputation
df_returns["log_date"] = pd.to_datetime(df_returns["log_date"])

median_refund = df_returns["refund_usd"].median() # 85.00
df_returns["refund_usd"] = df_returns["refund_usd"].fillna(median_refund)

print("\nRemaining Missing Values:\n", df_returns.isnull().sum()) # All 0
print("\nCleaned DataFrame:\n", df_returns)

# Task 4: Impact Evaluation
total_refund_after = df_returns["refund_usd"].sum()
print(f"\nTotal Refund Value After Imputation: ${total_refund_after:.2f}")
# $809.00
```

--- TASK 4 BUSINESS MEMO SOLUTION KEY ---

"To: Rachel Torres, Customer Success Lead Re: Customer Returns Data Quality Audit

The customer returns log was sanitized, eliminating 1 exact duplicate record and standardizing chaotic category text into three clear segments: Apparel, Electronics, and Home. Three missing refund values were imputed using the $85.00 median refund rate, preserving record integrity and bringing total audited refunds to $809.00. Median imputation was selected over row deletion to avoid losing valuable customer rating data associated with those return events."

2- **HOMEWORK SOLUTION**

```Python

```
Homework Problem 1 & 2 Python Solution
import pandas as pd
import numpy as np

supplier_log_data = {
    "order_id": ["SUP-901", "SUP-902", "SUP-903", "SUP-904", "SUP-904",
"SUP-905", "SUP-906", "SUP-907", "SUP-908", "SUP-909"],
    "order_date": ["2026-09-15", "2026-09-15", "2026-09-16", "2026-09-17",
"2026-09-17", "2026-09-18", "2026-09-18", "2026-09-19", "2026-09-20",
"2026-09-20"],
    "supplier_name": [" apex logistics ", "Global Freight", "APEX
LOGISTICS", "FastShip Inc", "FastShip Inc", " global freight ", "Apex
Logistics", "FastShip Inc", "GLOBAL FREIGHT", "FastShip Inc"],
    "unit_cost_usd": [120.00, 350.00, np.nan, 85.00, 85.00, 410.00, 115.00,
np.nan, 390.00, 90.00],
    "quantity_ordered": [50, 20, 40, 100, 100, 15, 60, 80, 25, 110]
}

df_supplier = pd.DataFrame(supplier_log_data)

Problem 1: Audit & Deduplicate
print("Initial Shape:", df_supplier.shape) # (10, 5)
df_supplier = df_supplier.drop_duplicates().reset_index(drop=True) # (9, 5)

Problem 2: String Cleaning, Dates & Imputation
df_supplier["supplier_name"] =
df_supplier["supplier_name"].str.strip().str.title()
df_supplier["order_date"] = pd.to_datetime(df_supplier["order_date"])

median_unit_cost = df_supplier["unit_cost_usd"].median() # 120.00
df_supplier["unit_cost_usd"] =
df_supplier["unit_cost_usd"].fillna(median_unit_cost)

df_supplier["total_order_cost"] = df_supplier["unit_cost_usd"] *
df_supplier["quantity_ordered"]
total_procurement_spend = df_supplier["total_order_cost"].sum()

print("Cleaned Unique Suppliers:", df_supplier["supplier_name"].unique()) **#
['Apex Logistics', 'Global Freight', 'Fastship Inc']**
print("Total Spend: $", total_procurement_spend) # $57,000.00
```
```

- **Homework Grading Criteria:**
  - Correctly removes 1 duplicate row (`SUP-904` duplicated).
  - Standardizes supplier names into 3 distinct clean entities: `Apex Logistics`, `Global Freight`, and `Fastship Inc`.
  - Imputes missing `unit_cost_usd` with $120.00 median.
  - Calculates total procurement spend as $57,000.00.